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Early Triassic paleomagnetic data from the Dolomites (Italy)

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Abstract. Prefolding high unblocking temperature magnetization components have been resolved from 28 sites of red siltstones from the Lower Triassic Werfen Formation in the Dolomites (Southern Alps). Site mean declinations are more scattered than site mean inclinations, indicating small-scale relative rotation among thrust sheets, particularly in the eastern part of the Dolomites affected by "Dinaric" (Paleogene) thrusting. Two sites from close to the Valsugana Line indicate large-scale (~80°) anticlockwise rotation in this zone of Alpine sinistral transpression. The overall mean direction (number of sites (N)=26, Declination = 331°, Inclination = 13°, α_{95} = 6.6°), excluding sites close to the Valsugana Line, gives a pole position (Latitude = 42°N, Longitude = 233°E, dp/dm = 3.4°/6.8°) which lies close to published Early Permian poles from the Southern Alps and to Permian and Early Triassic poles for Africa, implying very little net rotation of the Dolomites relative to Africa since Permian time.

INTRODUCTION

The Dolomites are part of the Southern Alps, the backthrust (south vergent) belt of the Alpine orogen. During the Jurassic-Cretaceous, the Southern Alps were located at the northern margin of Adria on the southern side of the Mesozoic Tethyan ocean. The Dolomites are located between the dextrally transpressive Pusteria Line to the north (a segment of the Insubric Line) and the SSE vergent Valsugana Thrust to the south (Figure 1). These two main fault zones have been active during Neogene time. The Valsugana Line marks the boundary between the Dolomites and the Venetian Prealps to the south. The Dolomites are relatively undeformed, possibly because of the presence of a structural high in the basement (Trento Horst) derived from Early Mesozoic development of the continental margin.

Paleomagnetic data from the Venetian Prealps have indicated no significant relative rotation among thrust sheets and a general concordance of Late Jurassic and Cretaceous data with the African apparent polar wander path [Channell et al., 1992a]. The lack of significant thrust sheet rotation is attributed to basement involvement and the deep-rooted nature of thrusting.

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Permian paleomagnetic data are available from several locations in the Southern Alps [Zijderveld et al., 1970; Manzoni, 1970; Heiniger, 1979], whereas Triassic paleomagnetic data from the Southern Alps are restricted to 12 sites from Middle and Late Triassic dykes and volcanic rocks [Manzoni, 1970]. Our objective in this study is to add to the Triassic paleomagnetic data set for the Southern Alps, investigate tectonic rotations in the Dolomites, and relate the paleomagnetic data to those from the Venetian Prealps and Africa.

The generally coarse-grained Upper Permian Valgardena Sandstone was found to behave unstably during thermal demagnetization, as was also noted by Manzoni [1970]. Our main sampling was in two Scythian red siltstone units (Siusi and Campil members) of the Werfen Formation.

Jurassic and Cretaceous paleomagnetic data from the Dolomites would permit comparison with published data of the same age from the Venetian Prealps (south of the Valsugana Line) [Channell et al., 1992a]. Unfortunately, suitable Jurassic-Cretaceous rocks crop out at very few locations in the Dolomites. Middle and Upper Jurassic limestones of the Ammonitico Rosso Formation crop out near the summit of Piz Boé (5 km WNW of Arabba, Figure 1), however, these sediments are located close to the Sella Thrust [Doglioni, 1990] and have been remagnetized (unpublished data). At Ra Stua (Figure 1), Upper Jurassic Ammonitico Rosso limestones are overlain by Lower Cretaceous Puez Marls, and two sites from this location yielded good quality paleomagnetic data.

GEOLOGICAL SETTING

Late Permian to Early Cretaceous crustal extension is well documented in the Southern Alps [Bernoulli et al., 1979]. The Early Liassic to Late Dogger sinistral transtensional tectonics in this region are considered to be the manifestation of the continental rifting phase which preceded the Dogger-Malm onset of oceanic rifting between Adria and Europe [Trümpy, 1982; Weissert and Bernoulli, 1985]. Inversion of extension to compression in the Southern Alps is apparent from middle Cretaceous time, coincident with the opening of the North Atlantic and the general change in relative motion of Africa and Europe from sinistral transtension to dextral transpression. Dextral transpression has been documented along the Insubric Line from Late Cretaceous time [Laubscher, 1983; Schmid et al., 1989].

The Dolomites constitute the hanging wall of the Valsugana Thrust. Differential subsidence in the Dolomites is evident

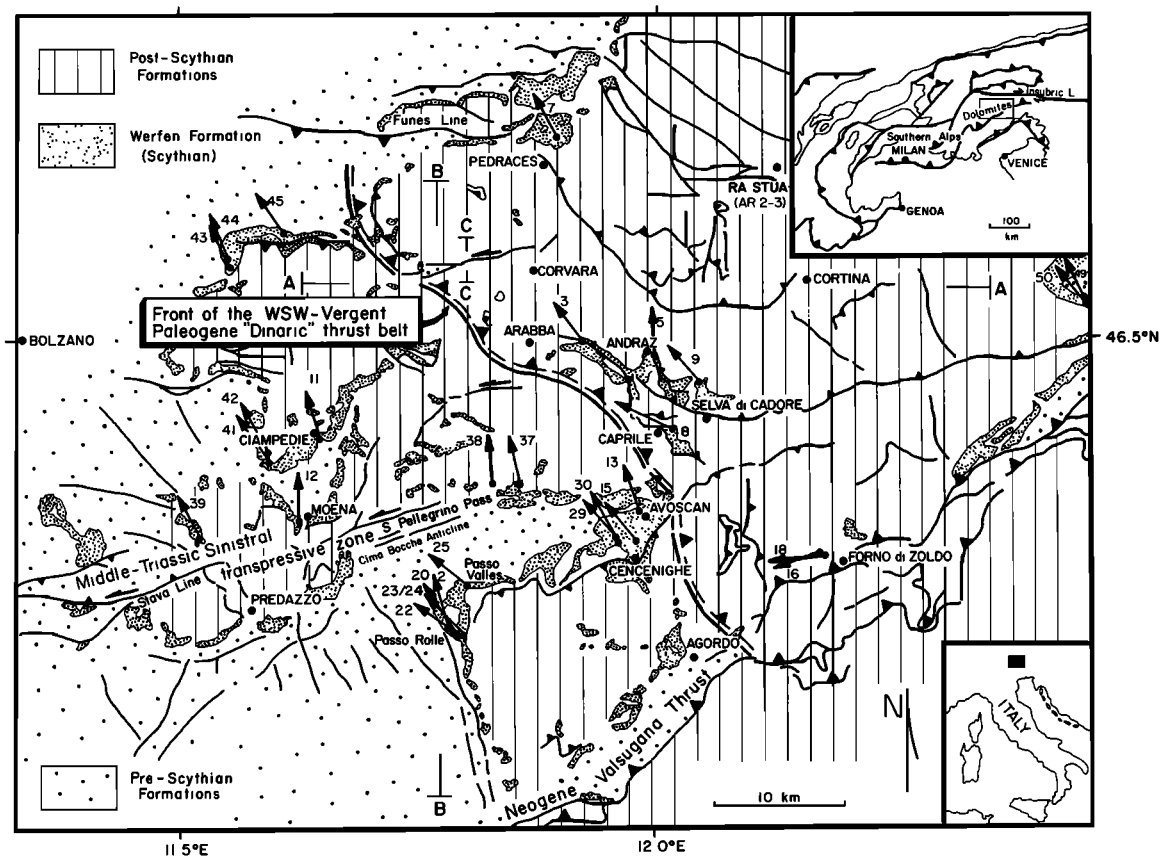


Figure 1. Simplified tectonic map of the Dolomites. Paleomagnetic sampling sites identified by site numbers (see Table 1) with arrows indicating site mean paleomagnetic declinations. The "Dinaric" (Paleogene) WSW vergent thrust front (dashed line with triangles) cuts through the region in a NW-SE direction. A-A, B-B and C-C mark the traces of cross sections in Figures 2a, 2b and 3, respectively.

from the general geologic map as the crystalline basement is at a higher stratigraphic level in the western part (Figure 1). Mesozoic extension generated mainly N-S trending normal faults (Figure 2a) which separate the Trento Horst to the west from the Belluno Graben to the east [Doglioni, 1992]. Strike-slip transfer faults trending N70°-90°E accommodated variations in E-W extension. Some of these transfer faults are reactivated Middle Triassic (late Ladinian) sinistral transpressive faults, the most important of which is the N70°E fault zone defined by the Stava Line and the northern limb of the Cima Bocche Anticline (Figures 1 and 2b). Along this trend, flower structures and high-angle strike-slip faults cut the basement and sedimentary cover, up to Ladinian strata. These features are crosscut by the late Ladinian-early Carnian intrusions of Predazzo and Monzoni, supporting the Middle Triassic age of this deformation [Doglioni, 1984]. Middle Triassic volcanic rocks along the tectonic line are shoshonitic in character [Lucchini et al., 1982] and have been attributed to a strike-slip tectonic setting [Sloman, 1989]. Similarly oriented Middle Triassic structures occur to the north along the Fedaia Pass and the Gardena Pass, where kinematic and stratigraphic data indicate a late Ladinian-early Carnian age of deformation. At the Gardena Pass (Figure 3), the left-lateral synsedimentary fault zone is recognized by striated fault planes, flower structures, different thicknesses of the Lower-Middle Triassic sequence on the two sides of the fault zone, and a syncline in the southern wall of the fault zone which involves the Lower Cassian Dolomite but not the Upper Cassian Dolomite.

The first post-Triassic phase of shortening in the Dolomites is Paleogene in age and is represented by WSW vergent thrusts of the "Dinaric" phase [Doglioni, 1987]. The front of this thrust belt runs from NW to SE across the Dolomites (Figure 1). The sedimentary cover of the Dolomites was shortened by at least 10-15 km during this deformation phase. Thrust ramps developed at the margins of Triassic carbonate platforms, and at the hinge zone between the Trento Horst and the Belluno Graben (Figure 2a).

From late Oligocene to Recent, the Dolomites and the Venetian Prealps were affected by SSE vergent thrusting of the "southalpine" deformation phase which involved the basement. The minimum total shortening in the Dolomites and Venetian Prealps during this deformation phase was 40-50 km [Doglioni, 1987]. During this Neo-Alpine phase, the Dolomites assumed a pop-up geometry bounded by the Valsugana Thrust and its conjugate back thrust, the Funes Line (Figures 1 and 2b). The pop-up structure produced a broad

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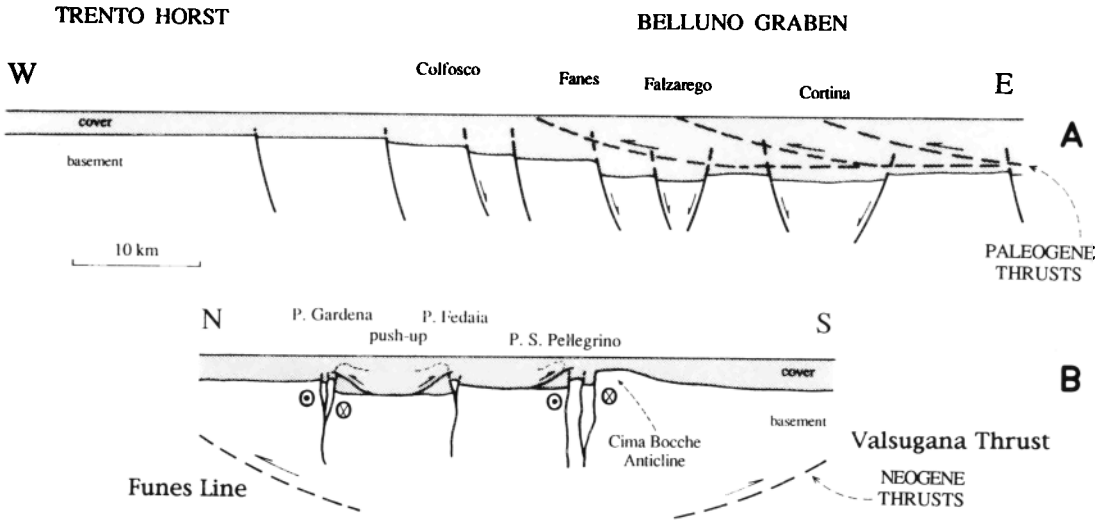


Figure 2. Schematic cross sections for the Dolomites at the end of the Mesozoic. The dashed lines indicate the later Paleogene traces of the WSW vergent "Dinaric" thrusts (Figure 2a) and the later Neogene SSE vergent Valsugana thrust and N vergent Funes Line (Figure 2b). A-A and B-B in Figure 1 mark locations of sections.

basement syncline in which the sedimentary cover was accommodated by flexural slip and flexural shear. The inherited Mesozoic structures (normal faults and strike-slip zones and carbonate platform to basin transitions) behaved as inhomogeneities which controlled transfer zones and undulations in shortening.

The Werfen Formation (Figure 4) predates Triassic compressive or strike-slip tectonics. The Werfen Formation is a 100 to 500-m-thick sequence of marls, marly limestone, siltstone, oolitic limestone, and evaporitic dolomites, formed in a shallow water platform environment [Broglio Loriga et al., 1982]. The Werfen Formation is Scythian in age and lies between the

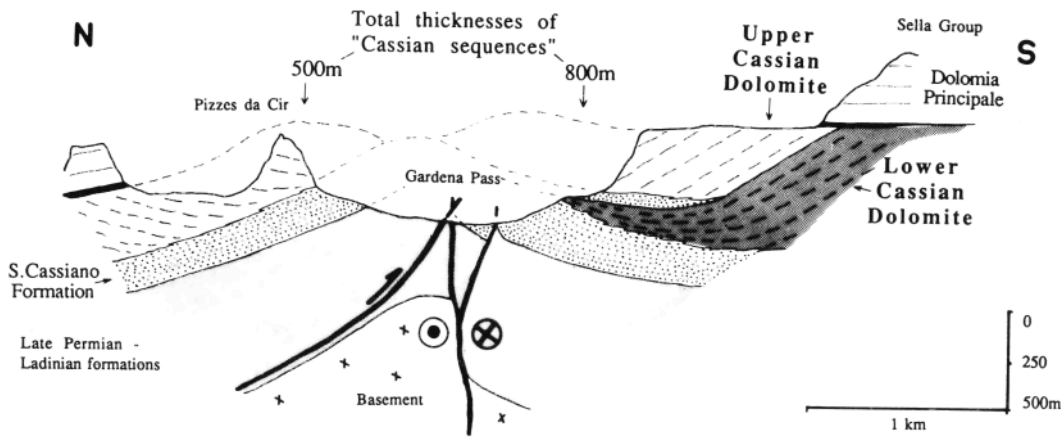


Figure 3. Section across the Gardena Pass illustrating stratigraphic evidence for Middle Triassic transtensional tectonics. Basement includes Lower Permian ignimbrites and Variscan crystalline rocks. C-C in Figure 1 marks location of section.

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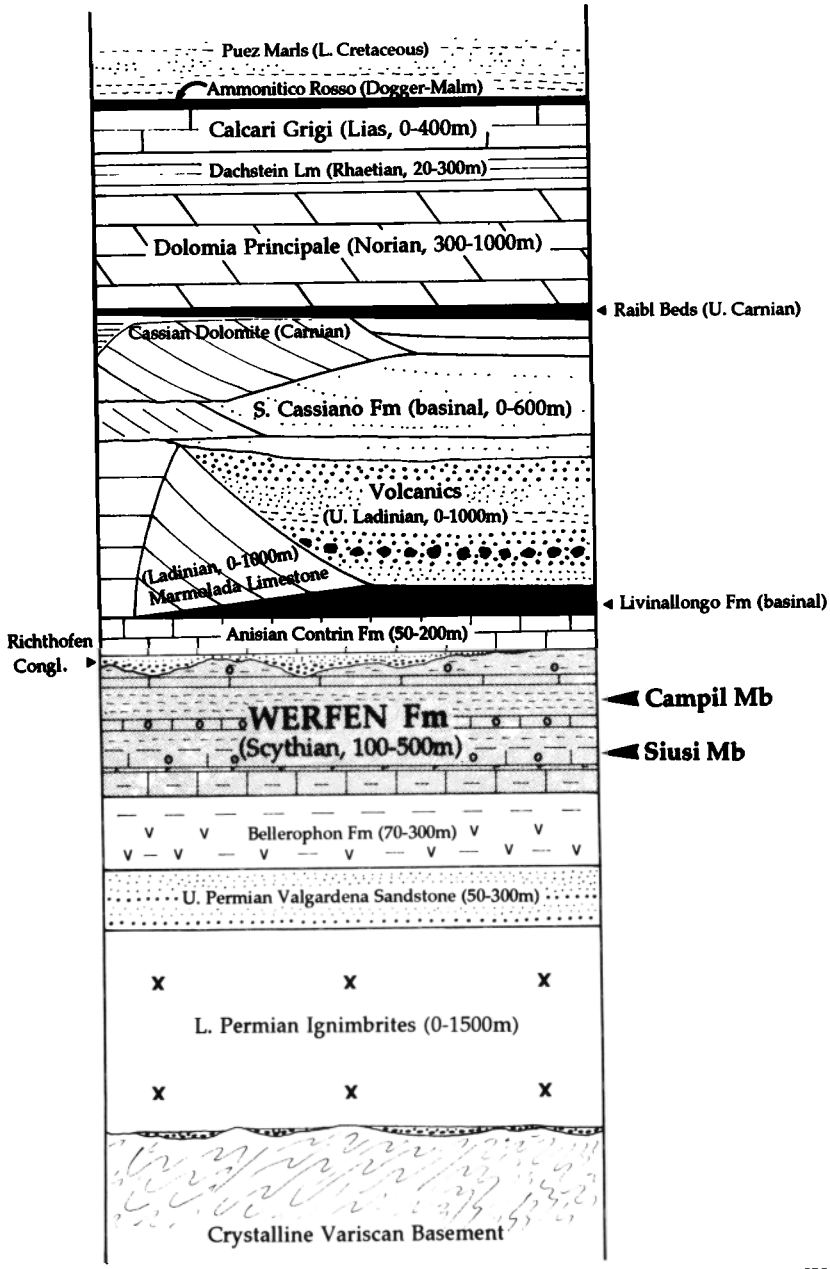


Figure 4. Generalized stratigraphy of the Dolomites (Campil and Siusi Members of the Werfen Formation, Ammonitico Rosso, and Puez Marls were sampled for paleomagnetic studies).

Upper Permian Bellerophon Formation and the upper Anisian Richthofen Conglomerate (Figure 4). The thickness variations of the Werfen Formation are due to synsedimentary tectonics which control the subsidence rate (lower rates to the west on the Trento Horst and higher rates to the east in the Belluno Basin) and to Anisian (transpressive?) tectonics which produced deep erosional scars in the Werfen Formation. The erosional features are overlain by the Richthofen Conglomerate, an upper Anisian alluvial facies [Dal Cin, 1967; Bosellini, 1968].

The Werfen Formation in the Dolomites comprises five or six depositional sequences with subtidal to supratidal repetitions, and has been subdivided into eight members and horizons [Broglio Loriga et al., 1982]. The middle part of the formation is represented by the Siusi Member (thickness of 90-100 m) which is overlain by the Gastropod Oolite Member (25-30 m) which in turn is overlain by the Campil Member (~100 m). The Siusi Member comprises grey marly mudstones, with carbonatic sand layers interpreted as storm layers, and red siltstones deposited in a subtidal environment. The Campil Member comprises red marly siltstones with sandy

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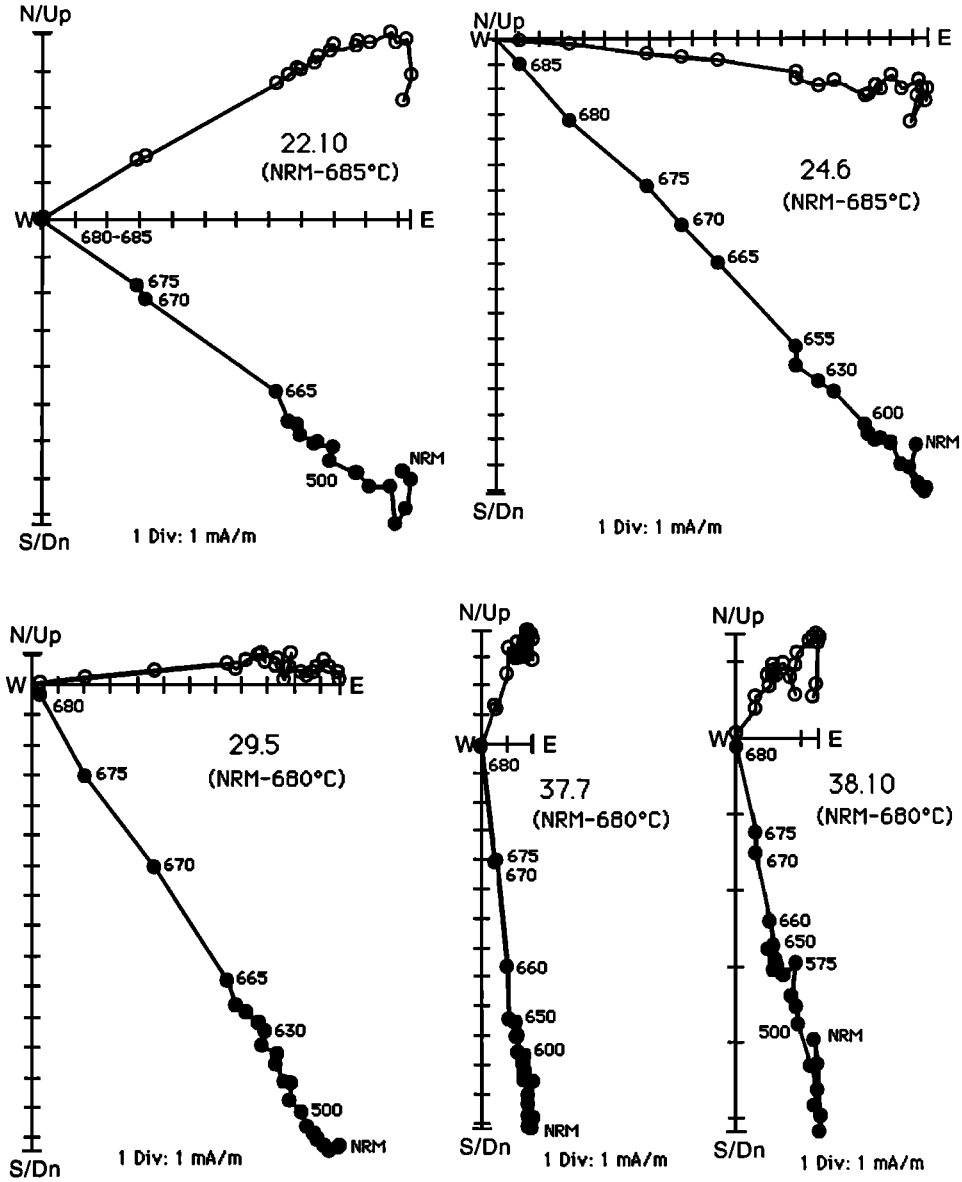


Figure 5. Orthogonal projections of tilt corrected thermal demagnetization data from the Campil Member of the Werfen Formation. Open and solid circles represent projections on the vertical and horizontal planes, respectively. The site number followed by the sample number, the temperature range, and temperatures for individual points (in degrees Celsius) are indicated.

storm layers and represents a depositional environment similar to that of the Siusi Member. The Campil Member is generally more terrigenous and less carbonatic than the Siusi Member.

The lower parts of the Siusi Member are characterized by a rich Claraia fauna, and the base of the Siusi Member corresponds to the Claraia clarai subzone which is correlated to the upper Griesbachian [Broglio Loriga et al., 1982]. The top of the Siusi Member and the entire Campil Member lie within the lower part of the Eumorphotis hinnitidea subzone which is correlated to the Dienerian-Smithian (Nammalian) interval [Broglio Loriga et al., 1982].

PALEOMAGNETIC ANALYSIS

Samples were collected with a gasoline-powered hand-held drill. Magnetization measurements were made on a 2G-755R cryogenic magnetometer housed in a magnetically shielded room at the University of Florida. All samples were progressively thermally demagnetized. Initial temperature increments were 50°C, and the increment was reduced to 5° or 10°C as the maximum unblocking temperature was approached. A single magnetization component was resolved from the red siltstones of the Siusi Member and Campil Member. The unblocking

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temperatures of this component are discrete, being largely restricted to the 650-680°C temperature range (Figure 5). Such unblocking temperature spectra are characteristic of the magnetization of fine-grained red clastic sediments [e.g, Kent and Opdyke, 1985; Miller and Kent, 1986; Channell et al., 1992b]. The high unblocking temperature magnetization component was resolved by picking (by eye) straight-line segments on orthogonal projections and determining the component directions associated with these straight-line segments, using the standard least squares method [Kirschvink, 1980]. Component directions for each sampling site were then combined using Fisher [1953] statistics (Table 1).

Magnetization components were resolved from 28 sites of a total of 34 sites collected from the Siusi and Campil members. Erratic demagnetization behavior appeared to correlate with coarser-grained lithologies. Two sites collected from the late Anisian Richthofen Conglomerate failed to yield stable magnetizations. Two sites (sites 16 and 18) from the Campil Member, located close to the Valsugana Thrust, are clearly rotated relative to the other sites. Sites 23 and 24 were collected from the two limbs of a fold and are treated as a single site in the analysis. The overall mean direction for the Campil and Siusi members is therefore based on 26 sites. Sites 16 and 18 are omitted from the overall mean.

TABLE 1. Site Mean Paleomagnetic Directions From the Dolomites

					Before Tilt		After Tilt			
Site	Location	Member	N/No	N/R	D	I	D	I	k	α_{95}
2	Passo Rolle	Campil	8/12	0/8	159.9	-13.1	161.9	-8.7	77.6	6.3
3	Arabba	Siusi	11/12	11/0	308.8	38.3	323.5	14.2	28.4	8.7
5	Andraz	Campil	11/12	11/0	342.6	42.2	357.5	15.4	13.1	13.1
7	Pedraces	Campil	9/14	0/9	153.6	11.4	153.6	-11.5	195.3	3.7
8	Caprile	Campil	16/16	0/16	115.0	-30.6	114.3	-16.2	31.3	6.7
9	San Cadore	Siusi	9/10	0/9	136.7	-3.2	138.8	-3.3	16.4	13.1
11	Ciampedie	Campil	10/10	0/10	162.4	-17.3	164.6	-3.6	30.6	8.9
12	Moena-Fassa	Campil	11/11	0/11	179.3	50.7	179.2	-6.2	135.1	3.9
13	Avoscan	Campil	8/13	0/8	160.9	30.1	160.9	-31.7	55.8	7.5
15	Avoscan	Campil	11/15	4/7	141.6	36.8	140.1	-24.2	20.5	10.3
16*	Forno Zoldo	Campil	12/12	0/12	66.8	29.2	66.9	-28.3	74.3	5.1
18*	Forno Zoldo	Campil	4/4	0/4	45.4	44.8	79.9	2.7	122.4	8.3
20	Passo Rolle	Campil	9/9	0/9	143.6	-6.2	146.5	-9.8	27.2	10.1
22	Passo Rolle	Campil	12/12	0/9	113.6	-28.0	125.2	-25.4	388.4	2.2
23/24†	Passo Rolle	Campil	18/19	0/18	157.7	-3.6			3.9	20.3
							141.1	4.0	51.9	4.8
25	Passo Valles	Campil	4/12	0/4	126.5	11.9	129.6	-1.9	154.1	7.4
29†	Cencenighe	Campil	17/19	1/16	140.0	4.4			13.2	10.2
							139.6	-5.9	102.6	3.5
30	Cencenighe	Campil	14/14	14/0	323.8	3.2	326.2	20.9	18.1	9.6
37	San Pellegrino	Campil	14/15	0/14	170.4	-41.2	169.6	-6.3	57.9	5.3
38	San Pellegrino	Campil	17/17	0/17	175.1	-40.7	175.4	-12.9	64.1	4.5
39	Pampeago	Campil	22/22	12/10	332.3	17.6	332.0	14.2	10.5	10.0
41	Paolina	Campil	14/14	0/14	147.0	13.5	145.9	12.4	62.8	5.1
42	Paolina	Campil	13/13	0/13	151.2	-11.7	152.0	-15.6	175.6	3.1
43	Castelrotto	Siusi	14/14	0/13	163.6	-36.1	159.9	-36.7	31.6	7.2
44	Castelrotto	Campil	14/14	0/14	158.1	-9.1	158.7	-16.7	182.3	3.0
45	Bulla	Campil	15/15	0/15	146.2	6.9	146.3	-5.7	78.8	4.3
49	Auronzo	Siusi	15/15	0/15	154.6	17.9	156.0	-15.2	55.8	5.2
50	Auronzo	Campil	14/14	0/14	161.7	-33.8	140.6	-19.9	10.9	12.6
AR2*	Ra Stua	A. Rosso	17/17	12/5	329.7	42.6	326.9	33.2	44.4	5.4
AR3*	Ra Stua	Puez Marl	15/15	15/0	338.9	48.6	335.6	37.1	136.2	3.3
Mean			26	4/22	329.7	8.4			7.8	10.8
							330.8	13.0	19.2	6.6
Mean			26	4/22		22.2			15.7	7.4
(inclination only)								13.9	45.5	4.3

N/No refers to the ratio of samples collected to samples yielding resolvable magnetization components. N/R refers to the ratio of normal to reversed polarity characteristic magnetization components. D and I refer to site mean declination and site mean inclination, respectively. k and α_{95} refer to statistical parameters after Fisher [1953]. Inclination only analysis is after Kono [1980].

Paleomagnetic pole latitude 41.7°N, longitude 232.5°E, dp/dm 3.4°/6.8°

* Not included in mean.

† Fold test.

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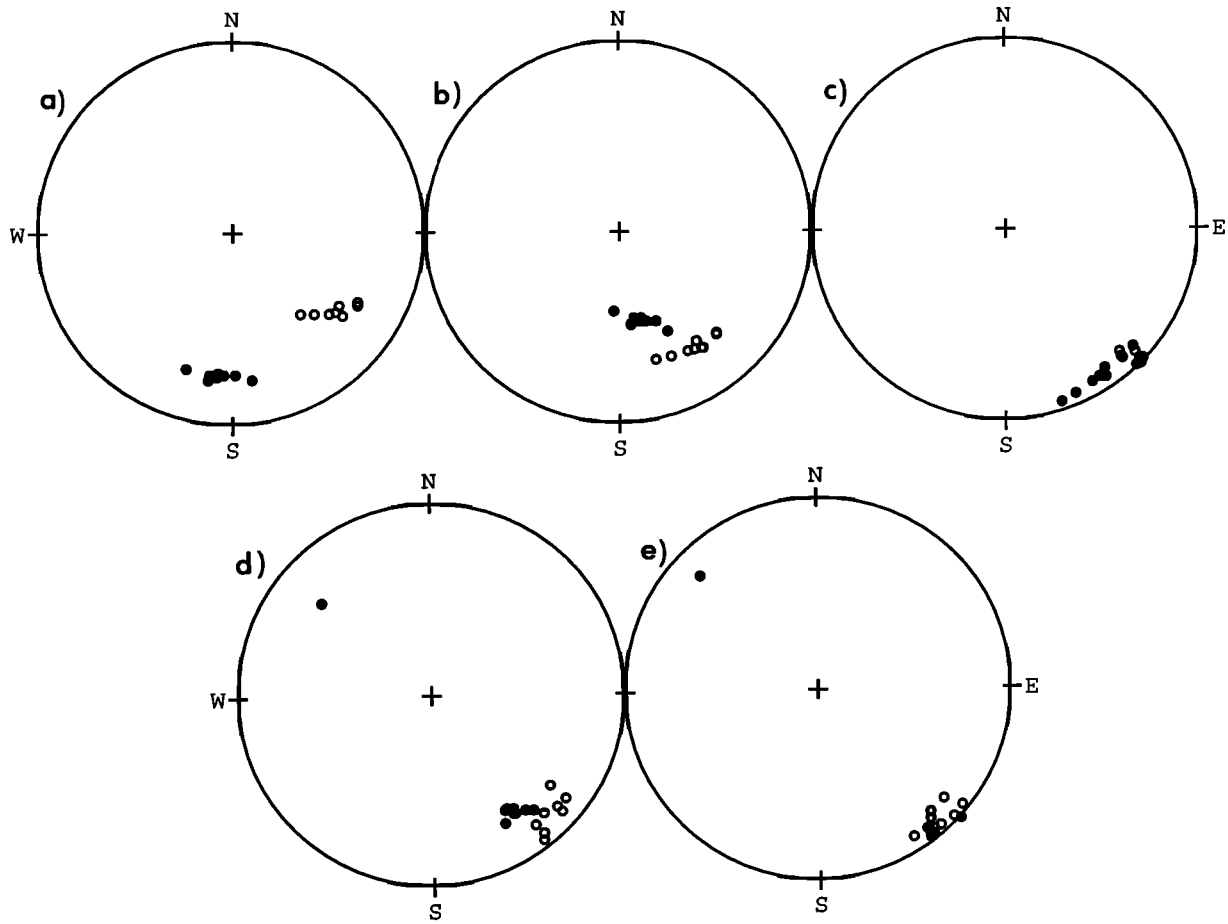


Figure 6. Fold tests from individual locations in the Campil member. (Top) Magnetization components from sites 23 and 24 with (a) no tilt correction (b) plunge correction only, and (c) plunge and limb tilt correction. (Bottom) Magnetization components from site 29 with (d) no tilt correction and (e) limb tilt correction (no plunge correction necessary). Solid and open circles indicate downward and upward inclinations, respectively. Equal-area projections.

At sites 23/24 and 29, local fold tests indicate that the high unblocking temperature magnetization predates Tertiary folding. At site 23/24, the fold plunges 33° to the NE, and the fold test involves a two-part tilt correction (Figures 6a-6c). The fold at site 29 is a gentle fold without plunge (Figures 6d-6e). Both local fold tests are positive at the 99% confidence level, using the criteria of McElhinny [1964] or McFadden [1990]. The 26 site mean component directions (Figure 7) are scattered, particularly in declination, reflecting the effects of local thrust sheet rotation. The site mean directions are, however, better grouped after tilt correction, and the improvement constitutes a positive regional fold test at the 99% confidence level, using the criteria of McElhinny [1964] or McFadden [1990]. The inclination-only method of Kono [1980] yielded almost the same mean inclination as that obtained using Fisher [1953] statistics applied to site mean directions (Table 1).

The overall mean paleomagnetic direction from the Campil and Siusi members of the Lower Triassic Werfen Formation (number of sites (N) = 26, Declination (D) = 331°, Inclination (I) = 13°, α_{95} = 6.6°) is similar to the southern Alpine mean directions obtained by Manzoni [1970] from the Upper Permian Valgardena Sandstone (N = 4, D = 328°, Inc = 13°, α_{95} = 18°) and Ladinian-Carnian dykes and volcanic rocks (N = 12, D = 329°, I = 25°, α_{95} = 9°), and to mean directions from Lower Permian volcanic rocks in the central part of the Southern Alps by Zijderveld et al. [1970] (N = 39, D = 330°, I = 20°, α_{95} = 5°) and Heiniger [1979] (N = 5, D = 320°, I = 14°, α_{95} = 8°).

Of the 28 site mean directions from the Campil and Siusi members (Table 1), 22 sites yielded reversed polarity magnetizations, three sites (sites 3, 5, 30) yielded only normal polarity components and three sites (sites 15, 29, 39) yielded both normal and reversed components. At the three mixed polarity sites, the normal polarity zone lies stratigraphically above the reversed polarity zone. Normal and reversed polarity zones are present in the Siusi Member and in the Campil Member. The predominance of reversed polarities at the Campil sites is con-

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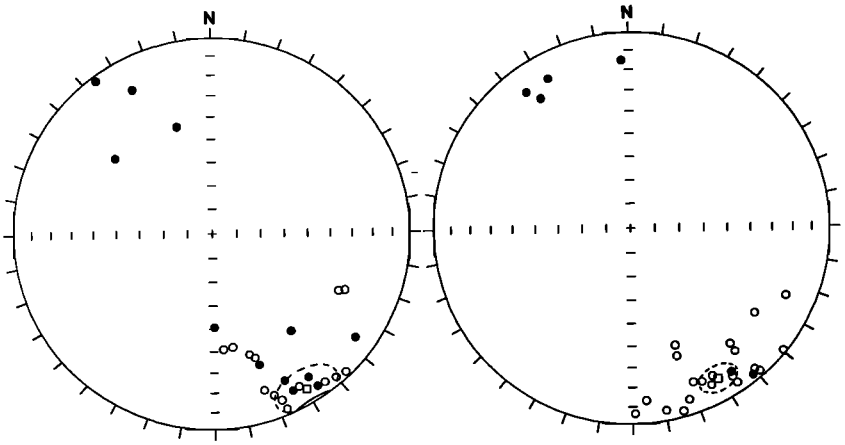


Figure 7. Site mean magnetization directions from 26 sites from the Campil and Siusi members of the Werfen Formation (left) before and (right) after structural tilt correction. Solid and open circles indicate downward and upward inclinations, respectively. The overall mean directions have upward inclinations and are indicated by squares with 95% confidence circles. Equal-area projections.

sistent with a reversed polarity bias in the Nammalian [see Molina-Garza et al., 1991].

TECTONIC ROTATIONS

Two sites (16 and 18 at Forno di Zoldo) have not been included in the overall mean direction for the Werfen Formation, as they are statistically distinct from the other site mean directions (Table 1). These two sites are located close to the Valsugana Thrust (Figure 1), and the site mean directions can be interpreted in terms of counterclockwise rotation (by about 80°) due to sinistral transpression along the Valsugana Thrust.

The sites to the south of the middle Triassic transpressive zone referred to as the Stava Line - Cima Bocche Anticline tectonic lineament (Figure 1) tend to be rotated by a few tens of degrees relative to those to the north. Middle Triassic sinistral motion along the fault zone may account for this observation, although later Alpine reactivation along the Stava Line - Cima Bocche Anticline fault zone or along other lines such as the Passo Rolle Line, an approximately N-S trending Mesozoic normal fault reactivated as a sinistral Alpine transfer fault, cannot be ruled out. The declination variation for the sites collected at Passo Rolle may be due to their close proximity to the Passo Rolle Line.

The mean directions for sites located within the WSW vergent Dinaric thrust belt are also scattered (Figure 1 and Table 1). The variable site mean declinations in this region may reflect the complicated pattern of structural undulations in Dinaric thrusting, which are probably due to frequent lateral and oblique thrust ramps generated by the strong facies and thickness variations in the Triassic strata.

It has been shown that Jurassic and Cretaceous paleomagnetic poles from the Venetian Prealps (south of the Valsugana Line) are very similar to coeval African poles [Channell et al., 1992a]. Jurassic and Cretaceous paleomagnetic data from the Dolomites would allow a comparison with the Venetian Prealps. Rocks of this age crop out at very few locations in the Dolomites, however, at Ra Stua (Figure 1), Upper Jurassic Ammonitico Rosso limestones are overlain by Lower Cretaceous Puez Marls (Figure 4). The site mean directions from these two formations (AR2 and AR3 in Table 1) indicate about 20° clockwise rotation relative to the mean directions from the Venetian Prealps [see Channell et al., 1992a]. The Ra Stua sites are, however, located in the hangingwall of a Dinaric WSW vergent thrust, inhibiting a general comparison between the Dolomites and the Venetian Prealps.

POLAR WANDER PATHS

The paucity of Permo-Triassic paleomagnetic data from Africa and other Gondwana continents results in poor definition of the Late Permian to Late Triassic portion of the African apparent polar wander path (APWP). Somewhat better estimates of the African APWP in this interval can be obtained by rotating North American and European paleomagnetic data to African coordinates, however, the choice of rotation parameters depends on the preferred Pangea configuration. We use the tight central Atlantic fit of Klitgord and Schouten (1986), which is consistent with the A2 Pangea configuration of Van der Voo (1993). In Figure 8, we plot (1) North American mean poles [Van der Voo, 1993] rotated to African coordinates using rotation parameters (Latitude = 66.95°, Longitude = -12.02°, Ω = 75.55°) from Klitgord and Schouten [1986]; (2) European mean poles [Van der Voo, 1993] rotated to African coordinates by rotation parameters (Latitude = 53.4°, Longitude = 1.5°, Ω = 55.7°) derived from Klitgord and Schouten [1986] and Srivastava and Tapscott [1986]; and (3) the Early Permian pole and the Late Triassic/Early Jurassic pole for NW Africa compiled by Van der Voo [1993] from Gondwana paleomagnetic data using the rotation parameters of Lottes and Rowley [1990] (the intervening Permo-Triassic poles are too poorly defined).

Paleomagnetic data from Lower Permian porphyries in the eastern and western Southern Alps have been acquired by Zijderveld et al. [1970] and Heiniger [1979], respectively. The Zijderveld et al. [1970] pole from near Bolzano (Figure 1) is

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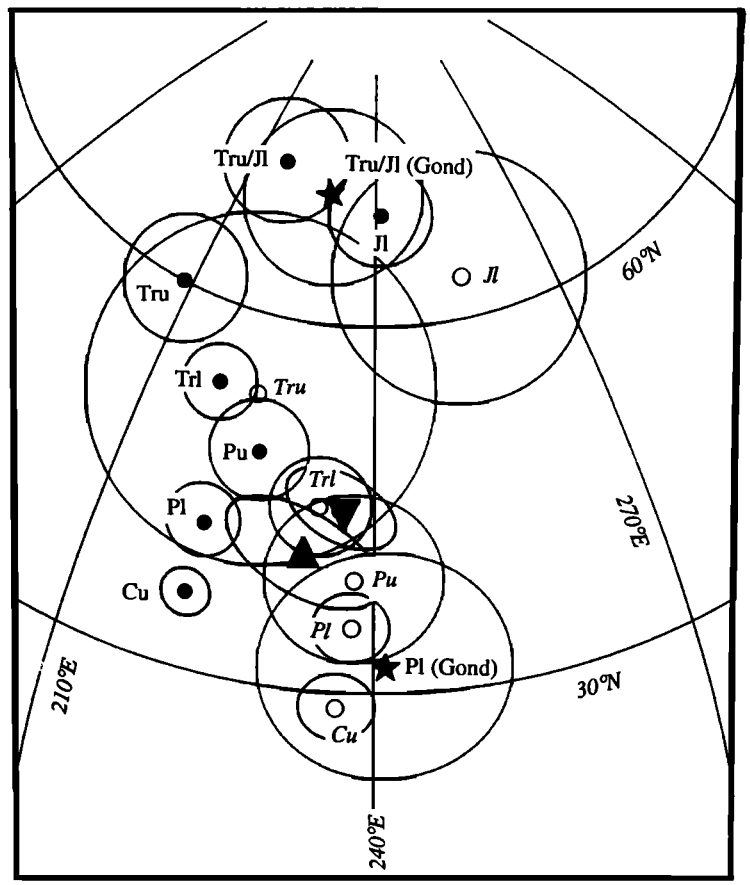


Figure 8. "African" paleomagnetic poles for Upper Carboniferous (Cu) to Lower Jurassic (Jl) compared to the poles from the Dolomites for Lower Triassic (upright triangle, this paper) and Lower Permian (inverted triangle, from Zijderveld et al., 1970). Solid circles: North American mean poles [Van der Voo, 1993] rotated to African coordinates by rotation parameters (Latitude = 66.95°, Longitude = -12.02°, Ω=75.55°) from Klitgord and Schouten [1986]. Open circles: European mean poles [Van der Voo, 1993] rotated to African coordinates by rotation parameters (Latitude = 53.4°, Longitude = 1.5°, Ω=55.7°) derived from Klitgord and Schouten [1986] and Srivastava and Tapscott [1986]. Solid stars: Early Permian and Late Triassic/Early Jurassic mean poles for NW Africa compiled by Van der Voo [1993] from Gondwana continents using reconstruction of Lottes and Rowley [1990] (intervening Permo-Triassic poles are too poorly defined).

the best defined Early Permian result from the Southern Alps, being based on 39 sites. The new Early Triassic pole from the Dolomites (42°N, 233°E) lies close to the Early Permian result from Bolzano (46°N, 236°E) (Figure 8).

The Early Triassic pole from the Dolomites lies close to Late Permian and Early Triassic poles on the rotated European APWP (Figure 8). Late Permian poles for Europe lie close to European Early Triassic poles and to a recently acquired Anisian (Middle Triassic) pole [see Théveniaut et al., 1992]. In contrast, the Early Triassic pole on the rotated North American APWP is significantly different from the Early Triassic pole from the Dolomites (Figure 8). The discrepancy may be attributed to longstanding problems in the reconstruction of Pangea and/or to errors in the North American APWP. The majority of the Early Triassic poles for North America are derived from the Colorado Plateau which has rotated clockwise relative to the North American craton. In the Van der Voo [1993] compilation, Colorado Plateau poles have been rotated counterclockwise by 5.4° to compensate for this tectonic rotation, however, a larger adjustment (13°-14°) now seems warranted [Kent and Witte, 1993]. The larger adjustment brings the rotated Early Triassic North American pole to lower latitudes closer to the Early Triassic pole from the Dolomites.

As mentioned above, Jurassic and Cretaceous paleomagnetic poles from the Southern Alps (mainly from the Venetian Prealps south of the Valsugana Line) are very similar to coeval African poles [Channell et al., 1992a]. Although Permo-Triassic poles for Africa are poorly defined, there is a general consistency between the Early Triassic pole from the Dolomites and coeval "African" poles (Figure 8), which leads

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us to the conclusion that there has been little net rotation of the Dolomites relative to the Venetian Prealps or to Africa since Permian time.

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